

1. 求方程 $(xy - x^3y^3)dx + (1 + x^2)dy = 0$ 满足条件 $y(0) = 1$ 的解.

2. 求方程 $x^2y'' - 3xy' + 4y = 0$ ($x > 0$) 的满足条件 $y(1) = 1, y'(1) = 1$ 的解, 其中

$$y' = \frac{dy}{dx}, y'' = \frac{d^2y}{dx^2}$$

$$\begin{aligned} 1. \quad P &= xy - x^3y^3, \quad Q = x^2 + 1 \\ \frac{\partial P}{\partial y} &= x - 3x^3y^2, \quad \frac{\partial Q}{\partial x} = 2x \\ (xy - x^3y^3)dx &= -(1 + x^2)dy \end{aligned}$$

$$y' = \frac{xy(x^2y^2 - 1)}{x^2 + 1} = \frac{x^3y^3}{x^2 + 1} - \frac{xy}{x^2 + 1}$$

$$\frac{1}{y^3} \frac{dy}{dx} = \frac{x^3}{x^2 + 1} - \frac{x}{y^2(x^2 + 1)}$$

$$-\frac{1}{2} \frac{dy^{-2}}{dx} = \frac{x^3}{x^2 + 1} - \frac{x}{x^2 + 1} \cdot y^{-2}$$

$$\text{令 } t = y^{-2}$$

$$\begin{aligned} \therefore -\frac{1}{2} \frac{dt}{dx} &= \frac{x^3}{x^2 + 1} - \frac{x}{x^2 + 1} \cdot t \\ \frac{dt}{dx} &= \frac{2x^3}{x^2 + 1} t - \frac{2x^3}{x^2 + 1} \end{aligned}$$

对应齐次方程为

$$\frac{dt}{dx} = \frac{2x}{x^2 + 1} t, \quad \frac{dt}{t} = \frac{2x}{x^2 + 1} dx, \quad \ln|t| = \ln|x^2 + 1| + C, \quad t = C(x^2 + 1)$$

$$\text{令 } C = \varphi(x),$$

$$t = \varphi(x)(x^2 + 1), \quad t' = \varphi'(x)(x^2 + 1) + \varphi(x) \cdot 2x$$

$$\therefore \varphi'(x) \cdot (x^2 + 1) = -\frac{2x^3}{x^2 + 1}$$

$$\varphi'(t) = -\frac{2x^3}{(x^2+1)^2} = -\int \frac{x^2}{(x^2+1)^2} dx^2 = \int \frac{\lambda}{(\lambda+1)^2} d\lambda = \int \frac{\lambda+1}{(\lambda+1)^2} d\lambda - \int \frac{1}{(\lambda+1)^2} d\lambda$$

$$\therefore \varphi(t) = -\ln|\lambda+1| - \frac{1}{\lambda+1} = \int \frac{1}{\lambda+1} d\lambda - \int \frac{1}{(\lambda+1)^2} d\lambda$$

$$\therefore t = -\left(\ln|x^2+1| + \frac{1}{x^2+1} + C\right) \cdot (x^2+1) = \ln|\lambda+1| + \frac{1}{\lambda+1}$$

$$\therefore y^2(x^2+1) \cdot \left(\ln|x^2+1| + \frac{1}{x^2+1} + C\right) + 1 = 0$$

$$y^2 \cdot [(x^2+1)\ln|x^2+1| + 1 + C(x^2+1)] + 1 = 0$$

$$x=0 \text{ 时, } y=1, \quad C=-2,$$

$$\therefore y^2 \cdot [2x^2 - (x^2+1)\ln|x^2+1| + 1] = 1$$

2. 求方程 $x^2 y'' - 3xy' + 4y = 0$ ($x > 0$) 的满足条件 $y(1) = 1$, $y'(1) = 1$ 的解, 其中

$$y' = \frac{dy}{dx}, y'' = \frac{d^2 y}{dx^2}$$

$$y'' - \frac{3}{x} y' + \frac{4}{x^2} y = 0$$

$$t = \ln x, \quad x = e^t$$

$$\frac{dy}{dx} = \frac{dy}{dt} \cdot \frac{dt}{dx} = \frac{1}{x} \frac{dy}{dt} = e^{-t} \frac{dy}{dt}$$

$$\frac{d^2 y}{dx^2} = \frac{d\left(\frac{dy}{dx}\right)}{dx} = \frac{d\left(\frac{dy}{dt}\right)}{dt} \cdot \frac{dt}{dx} = e^{-t} \cdot e^{-t} \left(\frac{d^2 y}{dt^2} - \frac{dy}{dt}\right) = e^{-2t} \left(\frac{d^2 y}{dt^2} - \frac{dy}{dt}\right)$$

$$\text{代入原式得} \quad \frac{d^2 y}{dt^2} - \frac{dy}{dt} - 3 \frac{dy}{dt} + 4y = 0$$

$$\therefore \frac{d^2 y}{dt^2} - 4 \frac{dy}{dt} + 4y = 0$$

$$\text{特征方程为} \quad \lambda^2 - 4\lambda + 4 = 0, \quad \lambda_1 = \lambda_2 = 2$$

$$\therefore y = (C_1 + C_2 t) \cdot e^{2t}$$

$$\text{即 } y = (C_1 + C_2 \ln x) \cdot x^2, y' = \frac{C_2}{x} \cdot x^2 + 2x(C_1 + C_2 \ln x) = C_2 x + 2C_1 x + 2C_2 x \ln x$$

$$\therefore \begin{cases} C_1 = 1 \\ 2C_1 + C_2 = 1 \end{cases} \Rightarrow \begin{cases} C_1 = 1 \\ C_2 = -1 \end{cases}$$

$$\therefore y = (1 - \ln x) \cdot x^2$$

3. 求方程 $y'' + y' - 2y = x + e^x + \sin x$ 的满足条件 $y(0) = -\frac{7}{20}$, $y'(0) = \frac{38}{15}$ 的解, 其中 $y' = \frac{dy}{dx}$, $y'' = \frac{d^2y}{dx^2}$.

特征方程为 $\lambda^2 + \lambda - 2 = 0$, $\lambda_1 = 1$, $\lambda_2 = -2$.

$\therefore$ 有对应齐次方程通解 $y = C_1 e^x + C_2 e^{-2x}$

① 对 $y'' + y' - 2y = x$, 0 不是特征根

设 $y = b_1 x + b_0$, $y' = b_1$, $y'' = 0$

$$b_1 - 2b_1 x - 2b_0 = -2b_1 x - 2b_0 + b_1 = x$$

$$\therefore \begin{cases} -2b_1 = 1 \\ b_1 - 2b_0 = 0 \end{cases} \therefore \begin{cases} b_1 = -\frac{1}{2} \\ b_0 = -\frac{1}{4} \end{cases}$$

$$\text{即 } y_1 = -\frac{1}{2}x - \frac{1}{4}$$

② 对 $y'' + y' - 2y = e^x$, 1 是单根

设 $y = ax \cdot e^x$, $y' = a(x+1)e^x$, $y'' = a(x+2)e^x$

$$\therefore 3ae^x = e^x, \quad a = \frac{1}{3}$$

$$y_2 = \frac{1}{3} x e^x$$

③ 对 $y'' + y' - 2y = \sin x$, $\pm i$ 不是特征根

设 $y = A \sin x + B \cos x$, $y' = A \cos x - B \sin x$, $y'' = -A \sin x - B \cos x$

$$\therefore (-2A-B-A)\sin x + (-2B+A-B)\cos x = \sin x$$

$$\therefore \begin{cases} -3A-B=-1 \\ A-3B=0 \end{cases} \therefore \begin{cases} A=-\frac{3}{10} \\ B=-\frac{1}{10} \end{cases}$$

$$\therefore y_3 = -\frac{3}{10}\sin x - \frac{1}{10}\cos x$$

则通解为 $y = C_1 e^x + C_2 e^{-2x} - \frac{1}{2}x - \frac{1}{4} + \frac{1}{3}x e^x - \frac{3}{10}\sin x - \frac{1}{10}\cos x$

$$y' = C_1 e^x - 2C_2 e^{-2x} - \frac{1}{2} + \frac{1}{3}(x+1)e^x - \frac{3}{10}\cos x + \frac{1}{10}\sin x$$

$$\begin{cases} C_1 + C_2 - \frac{1}{4} - \frac{1}{10} = -\frac{7}{20} \\ C_1 - 2C_2 + \frac{1}{3} - \frac{1}{2} - \frac{3}{10} = \frac{38}{15} \end{cases}$$

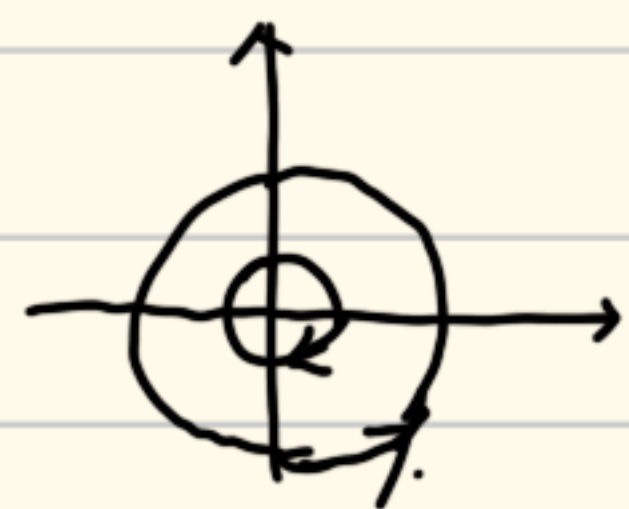
$$\text{即 } \begin{cases} C_1 + C_2 = 0 \\ C_1 - 2C_2 = 3 \end{cases} \therefore \begin{cases} C_1 = 1 \\ C_2 = -1 \end{cases}$$

$$\therefore y = e^x - e^{-2x} - \frac{1}{2}x - \frac{1}{4} + \frac{1}{3}x e^x - \frac{3}{10}\sin x - \frac{1}{10}\cos x$$

4. 设 $I(R) = \oint_{x^2+y^2=R^2} \frac{x dy - y dx}{(x^2+xy+y^2)^2}$, 证明 $\lim_{R \rightarrow +\infty} I(R) = 0$. 其中积分方向为逆时针方向.

直接用参数方程计算

$$(x + \frac{1}{2}y)^2 + \frac{3}{4}y^2 \leq \varepsilon^2$$



被积函数 $\varphi(x) = \frac{x dy - y dx}{(x^2+xy+y^2)^2}$ 在 $(0,0)$ 处无定义

记区域 $D: (x^2+xy+y^2) \leq \varepsilon^2, \varepsilon > 0$, $L: x^2+xy+y^2 = \varepsilon^2$, 逆时针方向, 记 $D_0: x^2+y^2 \leq R^2$

$L_0: x^2+y^2 = R^2$

$$P = \frac{-y}{(x^2+xy+y^2)^2}, \quad Q = \frac{x}{(x^2+xy+y^2)^2}$$

$$\frac{\partial P}{\partial y} = \frac{-(x^2+xy+y^2)^2 + y(2y+x) \cdot 2(x^2+xy+y^2)}{(x^2+xy+y^2)^4} = \frac{2y(2y+x) - (x^2+xy+y^2)}{(x^2+xy+y^2)^3} = \frac{y^2+xy-x^2}{(x^2+xy+y^2)^3}$$

$$\frac{\partial Q}{\partial x} = \frac{y^2+xy-x}{x^2+xy+y^2}$$

$$\frac{(x^2+xy+y^2)^2 - 2x \cdot (2x+y) \cdot (x^2+xy+y^2)}{(\quad)^4}$$

当 ε 充分小时, L 总在区域 D_0 内, $\therefore \oint_{L_0+L} \frac{x dy - y dx}{(x^2+xy+y^2)^2} = \iint_{D_0-D} (\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}) d\sigma \neq 0$

$$= \frac{(x^2+xy+y^2) - 4x^2 - 2xy}{(\quad)^3} = \frac{y^2-xy-3x^2}{(\quad)^3}$$

$$\therefore I(R) = \oint_L \frac{x dy - y dx}{(x^2 + xy + y^2)^2} = \frac{1}{\varepsilon^4} \oint_L x dy - y dx$$

$$x = \varepsilon \cos \theta, y = \varepsilon \sin \theta$$

$$dx = -\varepsilon \sin \theta, dy = \varepsilon \cos \theta$$

$$\text{原式} = \frac{1}{\varepsilon^4} \oint_L (\varepsilon^2 \cos^2 \theta + \varepsilon^2 \sin^2 \theta) d\theta$$

$$= \frac{1}{\varepsilon^4}$$

5. 设 L 为空间曲线 $\begin{cases} x^2 + y^2 = 1 \\ x + z = 1 \end{cases}$, 其正向为自 z 轴正向看下来的逆时针方向. 计算积分

$$I = \int_L (y - z + \sin^2 x) dx + (z - x + \sin^2 y) dy + (x - y + \sin^2 z) dz.$$



$$\frac{1}{\sqrt{2}} \iint_S \begin{vmatrix} 1 & 0 & 1 \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ y - z + \sin^2 x & z - x + \sin^2 y & x - y + \sin^2 z \end{vmatrix} dS = \frac{1}{\sqrt{2}} \iint_S (-1-1) + 0 + (-1-1) dS$$

$$= -2\sqrt{2} \iint_S dS$$

$$= -2\sqrt{2} \cdot \sqrt{2} \pi = -4\pi$$

6. 计算积分 $I = \iint_D (x + y + xy)^2 d\sigma$, 其中 $D = \{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2 \leq 1\}$.

7. 计算积分 $I = \iint_D \left(\frac{3x^2 \sin y}{y} + 2e^{x^2} \right) d\sigma$, 其中 D 由 $y = x, y = x^3$ 围成.



$$I = \iint_D (x^2 + y^2 + x^2 y^2 + 2x^2 y + 2xy + 2xy^2) d\sigma$$

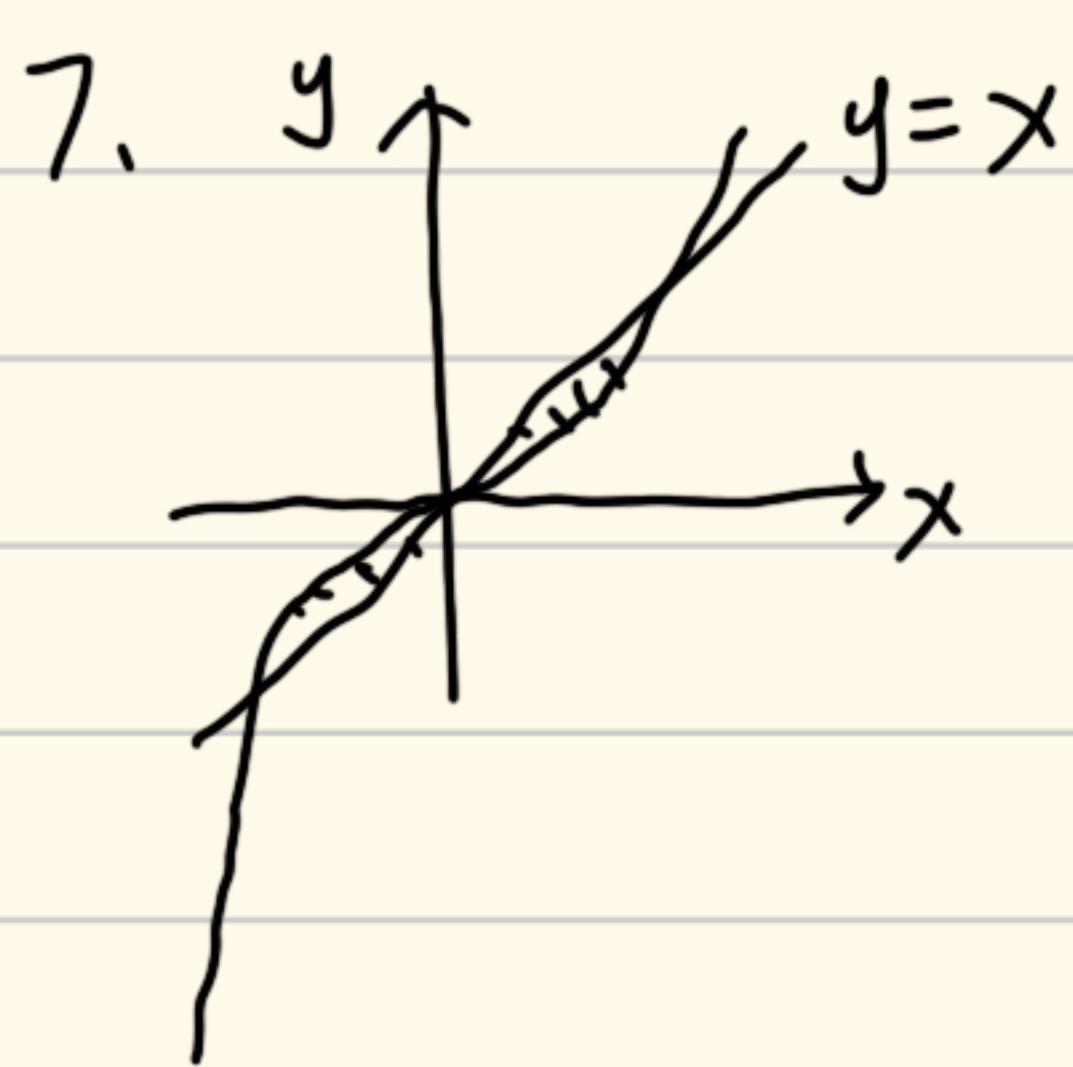
D 关于 x 轴, y 轴对称, 由对称性

$$I = \iint_D (x^2 + y^2 + x^2 y^2) d\sigma$$

$$x = r \cos \theta, y = r \sin \theta$$

$$I = \int_0^{2\pi} \int_0^1 r \cdot (r^2 + r^4 \sin^2 \theta \cos^2 \theta) dr d\theta = 4 \int_0^{2\pi} d\theta \int_0^1 (r^3 + r^5 \sin^2 \theta \cos^2 \theta) dr$$

$$= 4 \int_0^{2\pi} \left(\frac{1}{4} + \frac{1}{6} \sin^2 \theta \cos^2 \theta \right) d\theta$$



D 关于 $(0,0)$ 对称

而被积函数为 x, y 的偶函数

仅考虑第一象限

$$\begin{aligned} \iint_D \frac{3x^2 \sin y}{y} d\sigma &= 2 \int_0^1 dy \int_y^{\sqrt{y}} \frac{\sin y}{y} \cdot 3x^2 \cdot dy = 2 \int_0^1 (y - y^3) \frac{\sin y}{y} dy \\ &= 2 \int_0^1 (1 - y^2) \sin y dy \\ &= 2 \left[(y^2 - 3) \cos y - 2y \sin y \right] \Big|_0^1 \\ &= 2 \left[-2 \cos 1 - 2 \sin 1 + 3 \right] \\ &= 6 - 4 \cos 1 - 4 \sin 1 \end{aligned}$$

$$\begin{aligned} \iint_D 2e^{x^2} d\sigma &= 4 \int_0^1 dx \int_{x^3}^x e^{x^2} dy \\ &= 4 \int_0^1 (x - x^3) \cdot e^{x^2} dx \\ &= 2 \int_0^1 (1 - x^2) \cdot e^{x^2} dx^2 \\ &= 2 \left[e^x - (x^2 - 1) e^{x^2} \right] \Big|_0^1 \\ &= 2 \left[e - 1 + (-1) \right] = 2e - 4 \end{aligned}$$

$$\therefore \text{原式} = 2e + 2 - 4 \cos 1 - 4 \sin 1$$

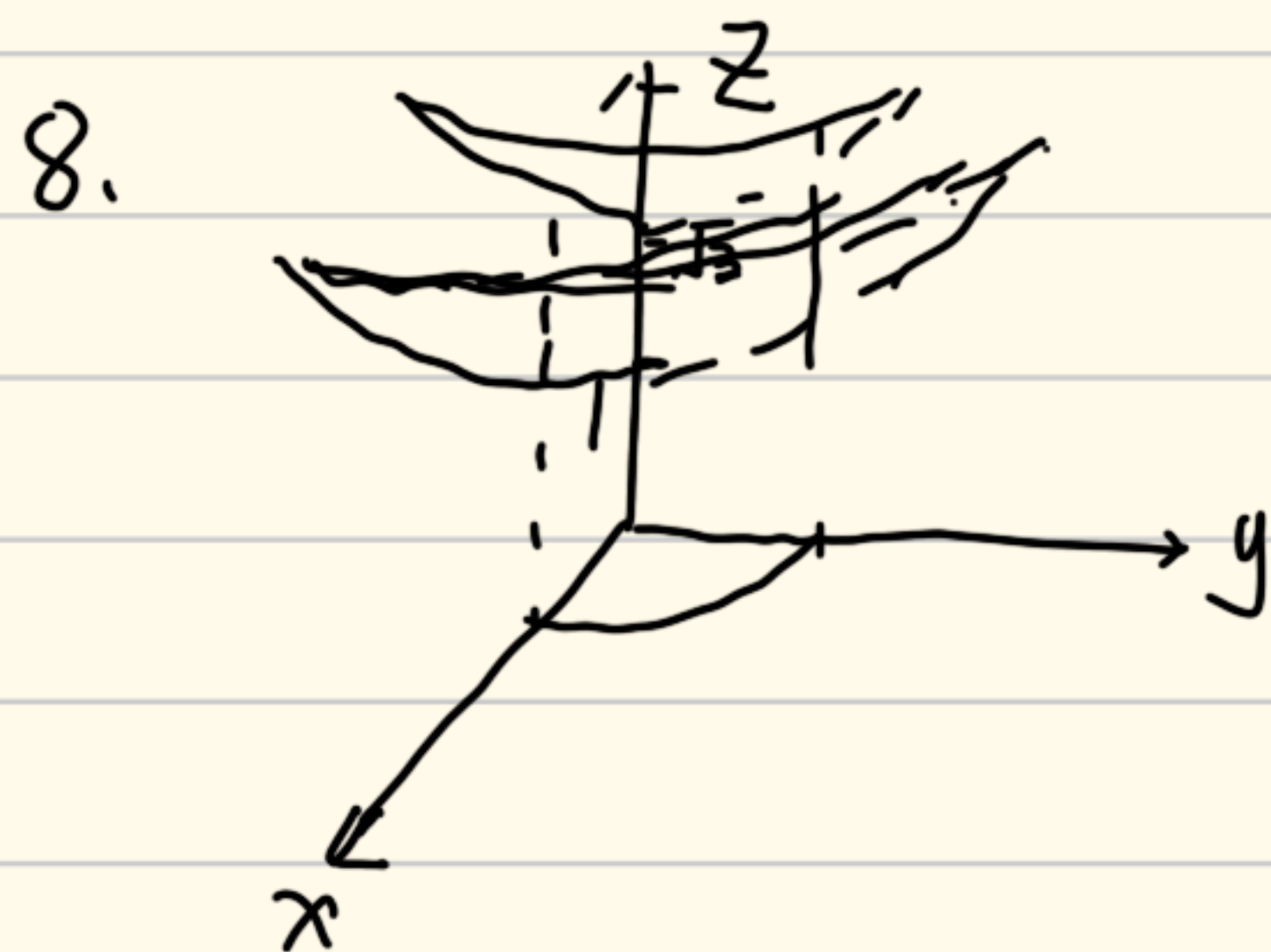
$$\begin{aligned} &= 4 \cdot \frac{2}{8} + \frac{1}{6} \cdot \frac{2}{16} \\ &= \frac{2}{2} + \frac{2}{24} = \frac{13}{24} 2 \end{aligned}$$

$$\begin{aligned} &\int (1 - y^2) \sin y dy \\ &= - \int (1 - y^2) d \cos y \\ &= - \left[(1 - y^2) \cos y - \int \cos y d(1 - y^2) \right] \\ &= - \left[(1 - y^2) \cos y + \int 2y \cos y dy \right] \\ &= - \left[(1 - y^2) \cos y + \int 2y d \sin y \right] \\ &= (y^2 - 1) \cos y - \left[2y \sin y - \int \sin y d2y \right] \\ &= (y^2 - 1) \cos y - 2y \sin y + 2 \int \sin y dy \\ &= (y^2 - 3) \cos y - 2y \sin y \end{aligned}$$

$x e^x$

8. 计算积分 $I = \iiint_{\Omega} \frac{(x+y+z)^2 \sqrt{1+x^2+y^2}}{(x^2+y^2+z^2)(1+x^2+y^2+z^2)} dv$, 其中 dv 即 $dx dy dz$, Ω 是由曲面 $z = \sqrt{1+x^2+y^2}$, $z = \sqrt{3(1+x^2+y^2)}$, $x^2+y^2 = 1$ 所围成的区域.

9. 计算积分 $I = \oint_{\Gamma} \left(\frac{y^2+y+4x^2}{4x^2+y^2} + \sin x^2 \right) dx + \left(\frac{4x^2-x+y^2}{4x^2+y^2} + \sin y^2 \right) dy$, 其中 Γ 是 $x^2+y^2 = 9 (y \geq 0)$, $\frac{x^2}{9} + \frac{y^2}{16} = 1 (y \leq 0)$ 所组成的闭曲线的逆时针方向.



$$x = r \cos \theta, y = r \sin \theta, z = z$$

关于 xOz , yOz 面对称, 只需计算第一象限

$$\begin{aligned} I &= \int_0^{2\pi} d\theta \int_0^1 r dr \int_{\sqrt{1+r^2}}^{\sqrt{3}\sqrt{1+r^2}} \frac{(r \cos \theta + r \sin \theta + z)^2 \sqrt{1+r^2}}{(r^2+z^2)(r^2+z^2+1)} dz \\ &= \int_0^1 r dr \int_{\sqrt{1+r^2}}^{\sqrt{3}\sqrt{1+r^2}} \frac{\sqrt{1+r^2}}{(r^2+z^2)(r^2+z^2+1)} dz \int_0^{2\pi} (r \cos \theta + r \sin \theta + z)^2 d\theta \\ &= 2\pi \int_0^1 r dr \cdot \int_{\sqrt{1+r^2}}^{\sqrt{3}\sqrt{1+r^2}} \frac{\sqrt{1+r^2} (r^2+z^2)}{(r^2+z^2)(r^2+z^2+1)} dz \\ &= 2\pi \int_0^1 r dr \cdot \int_{\sqrt{1+r^2}}^{\sqrt{3}\sqrt{1+r^2}} \frac{\sqrt{1+r^2}}{r^2+z^2+1} dz \end{aligned}$$

$$\begin{aligned} & r^2 \cos^2 \theta + r^2 \sin^2 \theta + z^2 \\ & + 2r^2 \sin \theta \cos \theta + 2rz \cos \theta + 2rz \sin \theta \\ & \frac{r^2 \sin^2 \theta}{r^2 \sin^2 \theta} \end{aligned}$$

$$\begin{aligned} z &= \sqrt{1+r^2} \cdot t, & z^2 &= (1+r^2) \cdot t^2 \\ \sqrt{1+r^2} dt &= 2\pi \int_0^1 r dr \int_1^{\sqrt{3}} \frac{\sqrt{1+r^2}}{1+r^2+(1+r^2)t^2} \sqrt{1+r^2} dt = 2\pi \int_0^1 r dr \int_1^{\sqrt{3}} \frac{1}{1+t^2} dt \\ &= 2\pi \cdot \frac{1}{2} \cdot \left(\frac{2}{3} - \frac{2}{4} \right) \\ &= \frac{2\pi}{12} \end{aligned}$$

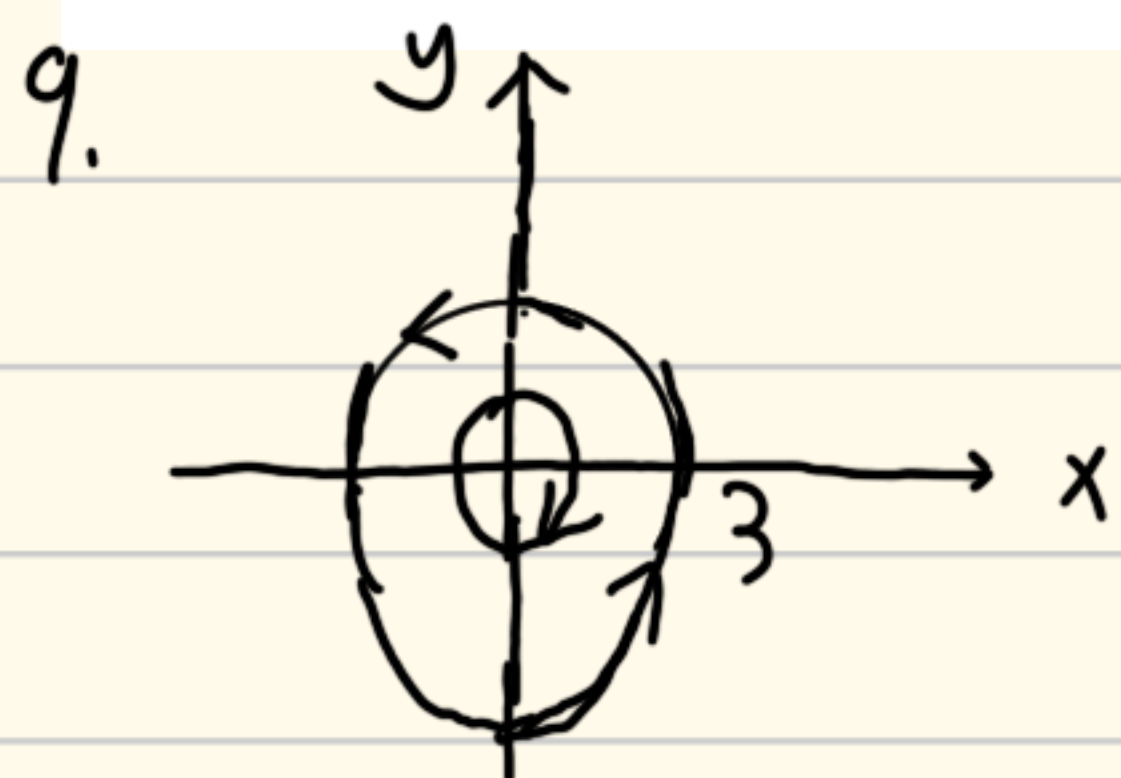
9. 计算积分 $I = \oint_{\Gamma} \left(\frac{y^2+y+4x^2}{4x^2+y^2} + \sin x^2 \right) dx + \left(\frac{4x^2-x+y^2}{4x^2+y^2} + \sin y^2 \right) dy$, 其中 Γ 是 $x^2 + y^2 = 9$ ($y \geq 0$), $\frac{x^2}{9} + \frac{y^2}{16} = 1$ ($y \leq 0$) 所组成的闭曲线的逆时针方向.

10. 设曲面 S 是柱体 $\Omega = \{(x, y, z) \mid x^2 + y^2 \leq 1, 0 \leq z \leq 1\}$ 的表面的外侧 (同时包含上表面、下表面、侧面). 计算下列积分:

(a) $I_1 = \iint_S (y-z)|x| dy dz + (z-x)|y| dz dx + (x-y)z dx dy$

(b) $I_2 = \iint_S (y-z)x^2 dy dz + (z-x)y^2 dz dx + (x-y)z^2 dx dy$

(c) $I_3 = \iint_S (y-z)x^3 dy dz + (z-x)y^3 dz dx + (x-y)z^3 dx dy$



$$P = \frac{y^2+y+4x^2}{4x^2+y^2} + \sin x^2$$

$$Q = \frac{4x^2-x+y^2}{4x^2+y^2} + \sin y^2$$

$$\frac{\partial P}{\partial y} = \frac{(2y+1)(4x^2+y^2) - (y^2+y+4x^2) \cdot 2y}{(4x^2+y^2)^2} = \frac{\cancel{8x^2y} + 2y^3 + 4x^2 + y^2 - 2y^3 - 2y^2 - \cancel{8x^2y}}{(4x^2+y^2)^2}$$

$$= \frac{4x^2 - y^2}{(4x^2+y^2)^2}$$

$$\frac{\partial Q}{\partial x} = \frac{(8x-1)(4x^2+y^2) - (4x^2-x+y^2) \cdot 8x}{(4x^2+y^2)^2} = \frac{\cancel{32x^3} - 4x^2 - y^2 + \cancel{8xy^2} - \cancel{32x^3} + 8x^2 - \cancel{8xy^2}}{(4x^2+y^2)^2}$$

$$= \frac{4x^2 - y^2}{(4x^2+y^2)^2}$$

$$\therefore \frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}$$

记 Γ_0^+ : $4x^2+y^2=\varepsilon^2$, ($\varepsilon>0$). 当 ε 充分小时, 一定包含在 Γ 内

$$\therefore \oint_{\Gamma_0^+} \left(\frac{y^2+y+4x^2}{4x^2+y^2} + \sin x^2 \right) dx + \left(\frac{4x^2-x+y^2}{4x^2+y^2} + \sin y^2 \right) dy = \iint_{D_0} \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) d\sigma = 0$$

$$\text{则 } I = \oint_{\Gamma_0^-} \left(\frac{y^2+y+4x^2}{4x^2+y^2} + \sin x^2 \right) dx + \left(\frac{4x^2-x+y^2}{4x^2+y^2} + \sin y^2 \right) dy$$

$$\frac{x^2}{\frac{1}{4}\epsilon^2} + \frac{y^2}{\frac{1}{2}\epsilon^2} = 1$$

$$= \frac{1}{\epsilon^2} \oint_{\Gamma} (y^2 + y + 4x + \epsilon^2 \sin x^2) dx + (4x^2 - x + y^2 + \epsilon^2 \sin y^2) dy$$

$$= \frac{1}{\epsilon^2} \iint_{D_0} (8x - 1 - 2y - 1) dG, \quad \# \# D_0: 4x^2 + y^2 \leq \epsilon^2$$

$$= \frac{2}{\epsilon^2} \iint_{D_0} (4x - y - 1) dG = \frac{2}{\epsilon^2} \iint_{D_0} (4x - y) dG - \frac{1}{2} \epsilon^2 \cdot 2 \cdot \frac{2}{\epsilon^2} = \frac{2}{\epsilon^2} \iint_{D_0} (4x - y) dG - 2$$

$$= \frac{2}{\epsilon^2} \int_0^{2\pi} d\theta \int_0^\epsilon (2r \cos \theta - r \sin \theta) \cdot \frac{1}{2} r \cdot dr - 2$$

$$= \frac{2}{\epsilon^2} \int_0^{2\pi} d\theta \int_0^\epsilon (2 \cos \theta - \sin \theta) \cdot \frac{1}{2} r^2 \cdot dr - 2$$

$$= \frac{1}{3} \epsilon \int_0^{2\pi} (2 \cos \theta - \sin \theta) \cdot d\theta - 2$$

$$= 0 - 2$$

$$= -2$$

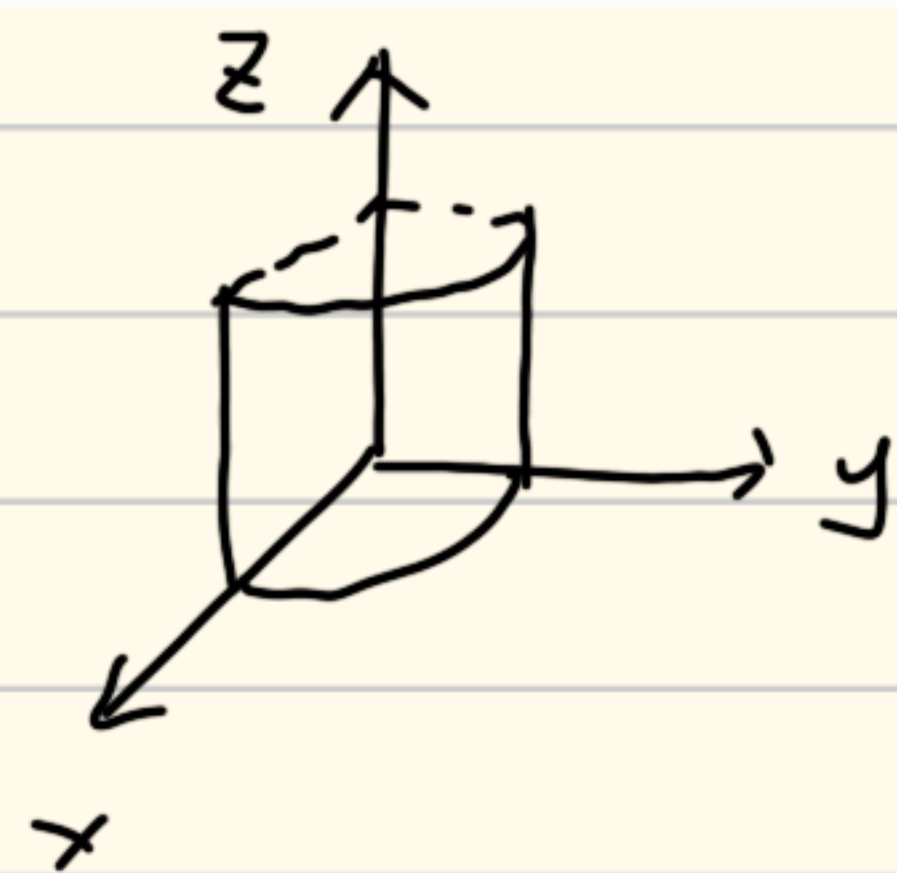
$$x = \frac{1}{2} r \cos \theta, \quad y = r \sin \theta$$

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(c) $I_3 = \iint_S (y-z)x^3 dy dz + (z-x)y^3 dz dx + (x-y)z^3 dx dy$



(a) 关于 x 和 z , y 和 z 面都对称, 只计算第一象限即可

$$I_1 = 4 \iiint_S (y-z) \cdot x dy dz + (z-x) y \cdot dz dx + (x-y) z dx dy$$

记 $S_1: 0 \leq x \leq 1, 0 \leq z \leq 1, y=0$

$S_2: 0 \leq y \leq 1, 0 \leq z \leq 1, x=0$

$$\text{则 } \frac{I_1}{4} = \iiint_V (y-z + z-x + x-y) dV - \iint_{S_1} 0 dz dx - \iint_{S_2} 0 dz dy = 0$$

$$\therefore I_1 = 0$$

(b) 由高斯公式: $I_2 = \iiint_S 2x(y-z) + 2y(z-x) + 2z(x-y) dV = 0$

(c) 0